Deep Learning

Experiments 16–20

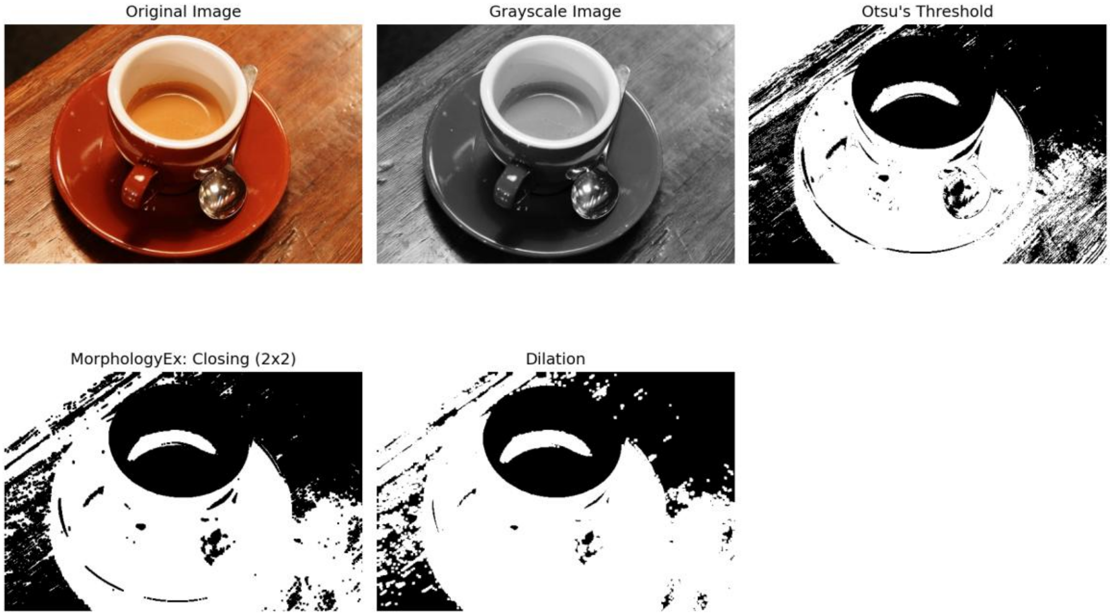
Experiment 16: Image Segmentation using Python

Aim: To demonstrate the segmentation of image using python

Program:

```
import numpy as np
import cv2
from matplotlib import pyplot as plt
from google.colab import drive

# Mount Google Drive if not already mounted
try:
    drive.mount('/content/drive')
except:
    pass  # Drive is already mounted

img = cv2.imread('/content/drive/MyDrive/dog.jpeg')

# Check if the image was loaded successfully
if img is None:
    print("Error: Could not load image. Please check the file path.")
else:
    b, g, r = cv2.split(img)
    rgb_img = cv2.merge([r, g, b])
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    ret, thresh = cv2.threshold(gray, 0, 255, cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU)
    kernel = np.ones((2, 2), np.uint8)
    closing = cv2.morphologyEx(thresh, cv2.MORPH_CLOSE, kernel, iterations=2)
    sure_bg = cv2.dilate(closing, kernel, iterations=3)

    plt.figure(figsize=(12, 8))
    plt.subplot(231); plt.imshow(rgb_img); plt.title("Original Image"); plt.axis('off')
    plt.subplot(232); plt.imshow(gray, 'gray'); plt.title("Grayscale Image");
plt.axis('off')
    plt.subplot(233); plt.imshow(thresh, 'gray'); plt.title("Otsu's Threshold");
plt.axis('off')
    plt.subplot(234); plt.imshow(closing, 'gray'); plt.title("MorphologyEx: Closing
(2x2)"); plt.axis('off')
    plt.subplot(235); plt.imshow(sure_bg, 'gray'); plt.title("Dilation");
plt.axis('off')
    plt.tight_layout()
    plt.show()
    plt.imsave('dilation.png', sure_bg)
```

Output:



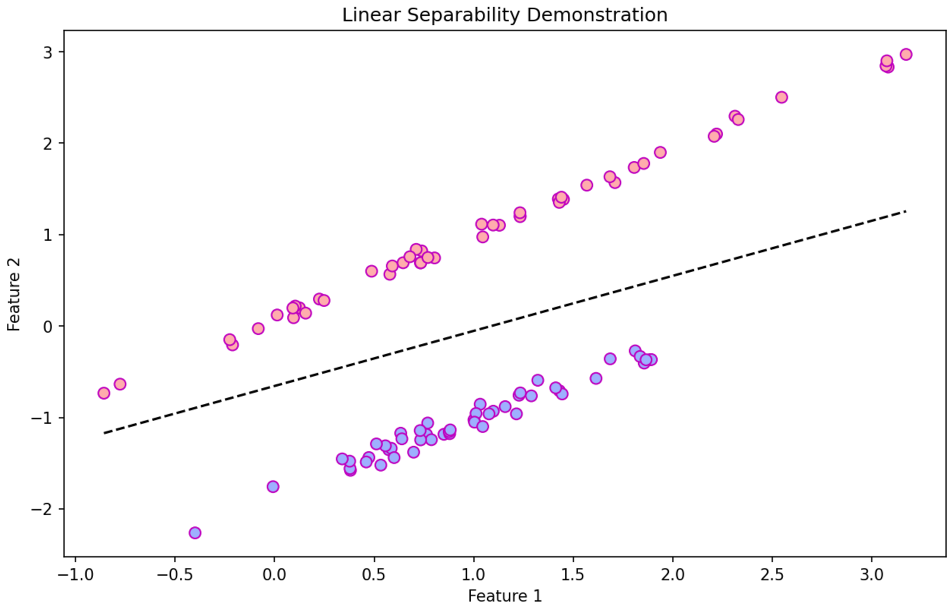
Experiment 17: Linear Separability using Python

Aim: To demonstrate linear separability using python code

Program:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score


X, y = make_classification(n_samples=100, n_features=2, n_informative=2,
                            n_redundant=0, n_clusters_per_class=1, random_state=42)


model = LogisticRegression()
model.fit(X, y)
y_pred = model.predict(X)
accuracy = accuracy_score(y, y_pred)
print(f"Accuracy: {accuracy:.2f}")


plt.figure(figsize=(10, 6))
plt.scatter(X[:, 0], X[:, 1], c=y, cmap='berlin', edgecolor='m', s=50)


coef = model.coef_[0]
intercept = model.intercept_
x_vals = np.linspace(X[:, 0].min(), X[:, 0].max(), 100)
y_vals = -(coef[0] * x_vals + intercept) / coef[1]


plt.plot(x_vals, y_vals, 'k--')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.title('Linear Separability Demonstration')
plt.show()
```

Output:

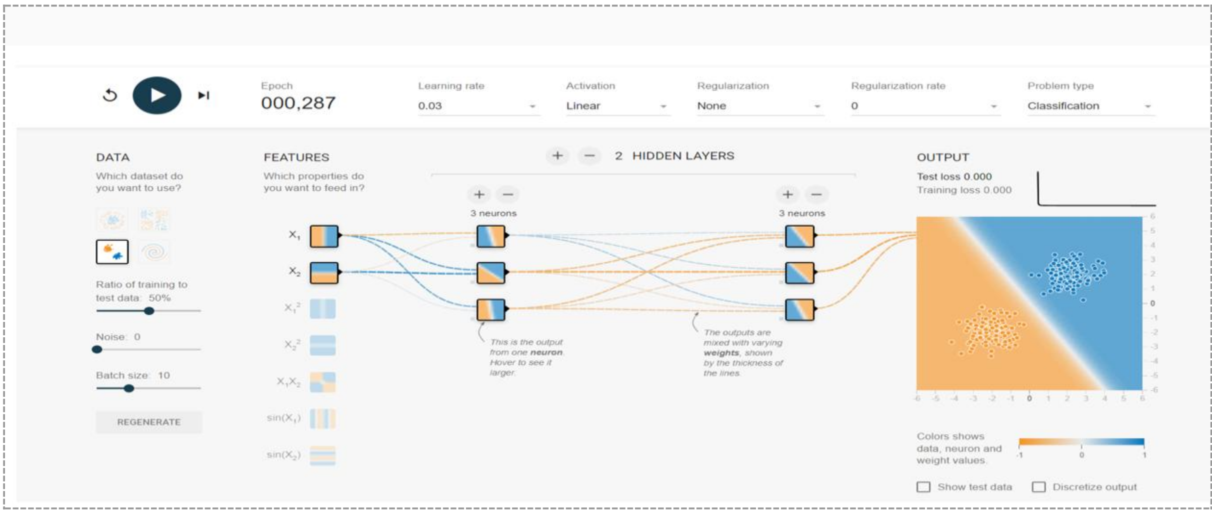


Experiment 18: Neural Network Analysis — Two Class, Linear Activation

Aim: Neural network analysis for Two class, Learning rate: 0.03, Activation: Linear, Hidden Layers: 02, and Hidden neurons: 03.

Dataset	Gaussian ("Two class")
Learning Rate	0.03
Activation	Linear
Hidden Layers	2
Neurons / Hidden Layer	3
Network Shape	3,3

Output:

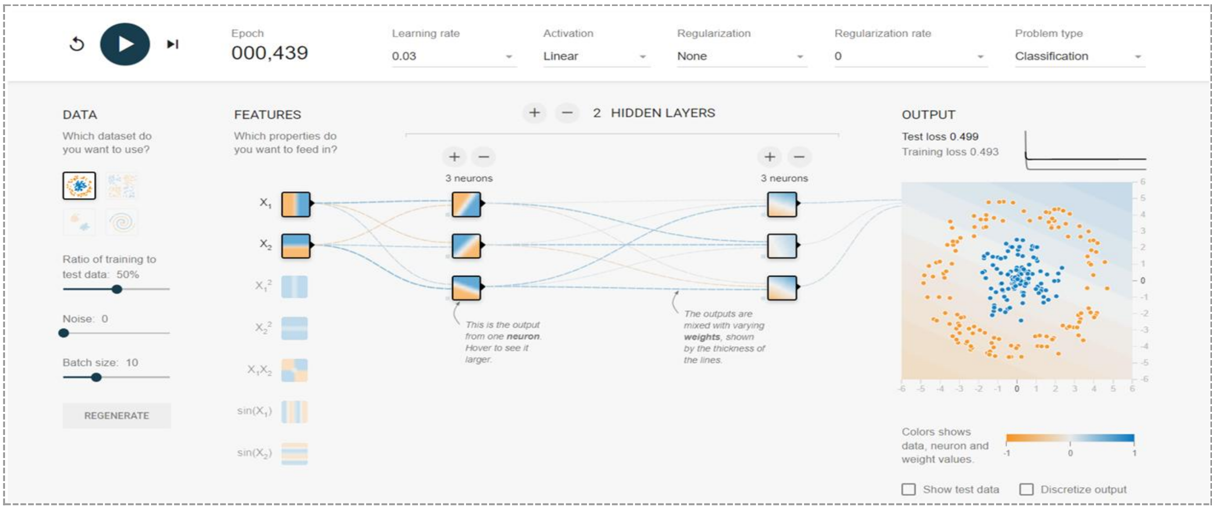


Experiment 19: Neural Network Analysis — Circular Data, Linear Activation

Aim: Neural network analysis for circular data class, Learning rate: 0.03, Activation: Linear, Hidden Layers: 02, and Hidden neurons: 03.

Dataset	Circle
Learning Rate	0.03
Activation	Linear
Hidden Layers	2
Neurons / Hidden Layer	3
Network Shape	3,3

Output:



Experiment 20: Neural Network Analysis — Multi Class, Linear Activation

Aim: Neural network analysis for Multi class, Learning rate: 0.01, Activation: Linear, Hidden Layers: 02, and Hidden neurons: 02.

Dataset	Exclusive OR ("Multi class")
Learning Rate	0.01
Activation	Linear
Hidden Layers	2
Neurons / Hidden Layer	2
Network Shape	2,2

Output: